Hypothetical routes from parataxis to hypotaxis

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Abstract

In many approaches to the evolution of complex syntactic structures from the 19th century onwards, it has been claimed that subordinate clauses develop from sequences of independent main clauses. In traditional terms, such a development has also been termed as a change from 'parataxis' into 'hypotaxis'. Hypothetical routes for this type of development have been discussed in the realms of complement, relative, and adverbial clauses. However, if one looks at the steps in the development of specific types of subordinate clauses in individual languages (Indo-European and beyond), the empirical support for the respective hypothetical routes is weak. Upon closer investigation, it has often turned out that the purported developments did not originate in paratactic structures, but in pre-existing subordinate structures with relative clauses.

Keywords

Subordination, parataxis, hypotaxis, complementizers, complement clauses, relative clauses, adverbial clauses

Main text

1. Introduction

The term 'parataxis' (from Greek *para-* 'beside' + *tassein* 'arrange') was arguably first used by Wilhelm Thiersch, who described parataxis as a simple juxtaposition of clauses and claimed that evidence from Homerian Greek suggested that this was the original way of clause combining (Thiersch 1826, 3rd edition). 'Syntaxis', which was later called 'hypotaxis' (from Greek *hypo-* 'beneath' + *tassein* 'arrange'), developed out of parataxis via the introduction of particles and certain "word forms" which turned into markers for clausal relations. Thiersch also aligns the paratactic clause combining with earlier stages of glottogenesis and associates the diachronic evolution of complex, hypotactic syntactic structure with increasing cognitive capacities of mankind (Dymarski 2014). Some philologists in the second half of the nineteenth century drew parallels between linguistic and biological evolution. Notably, August Schleicher compared languages to natural organisms independent of human influence and subject to the general biological laws of evolution. Under such a Darwinian perspective, originally more primitive languages (or language stages) with a simple juxtaposition of clauses were gradually replaced by languages (or stages) with more complex structures. However, it was not before the advent of the Neogrammarians that concrete developmental paths for different types of hypotactic clause combinations were put forward (e.g., Delbrück 1900, 445 ff.). Delbrück (1900, 413), in his monumental *Vergleichende Syntax der indogermanischen Sprachen,* goes even so far as to claim that “the claim that hypotaxis originated from parataxis has become common knowledge of science“ [ibid., our translation]. On the other hand, he argues against the common doctrine that persisted until 20th-century Indo-European linguistics, namely that Proto-Indo-European did not have any hypotactic structures. According to Delbrück, relative clauses and (adverbial) conjunctional clauses already existed in the common ancestor of the Indo-European languages (Delbrück 1900, 379, 415). The proposed pathways from paratactic structures are thus hypothetical routes in prehistoric times more than developments actually attested in the old Indo-European languages. Almost one hundred years later, Harris & Campbell (1995) on the basis of cross-linguistic studies also beyond Indo-European come to the conclusion that the empirical foundation for parataxis-to-hypotaxis processes is thin: “While most languages have parataxis, we have no direct evidence of it developing into hypotaxis” (Harris & Campbell 1995, 286). When evaluating research on the parataxis-to-hypotaxis hypothesis, a distinction should be made whether individual studies and approaches refer to the origin of hypotaxis, i.e., “to the first appearance of hypotaxis in a language”, or “to its repeated renewal” (Harris & Campbell 1995, 283). Renewal concerns the fact that in many languages with indisputable pre-existing hypotactic structures, new complementizers or other types of subjunctions evolve from lexemes that served other functions previously.

The view that hypotaxis originates in parataxis and that its development is associated with evolutionary progress in human communication has also been popular in 20th-century research, and is still today, notably in functionalist approaches (e.g., Givón 1979, Haspelmath 2010). In generative accounts, recursivity (including arguably recursive clause combining in the form of hypotaxis) is generally considered an essential and unique feature of the human language faculty (Hauser et al. 2002). Chomsky has flanked this conviction with arguments against a gradualist adaptionist evolutionary approach to syntax (Chomsky 2010, Berwick & Chomsky 2016). Weiß (2019) develops a theory (based on the minimalist concept of phasehood) that rules out parataxis-to-hypotaxis developments in principle.

In the 20th-century descriptive tradition of (mainly) the Indo-European languages, the term ‘subordination’ has come in use instead of hypotaxis. Just like in the case of ‘hypotaxis’ (and ‘parataxis’), there is no commonly agreed formal definition. There is some consensus, however, that a subordinate clause is dependent on a superordinate clause and thus cannot stand in isolation. In the grammatical tradition of the Indo-European languages, syntactic criteria such as the presence of complementizers (traditionally called subjunctions) or relative and interrogative pronouns, and word order characteristics play a major role along with grammatical mood (e.g., subjunctive). Subordination can be expressed by different types of finite clause types as well as by clause-like syntagms with non-finite verbal forms, such as infinitives, verbal nouns, and participles. Cross-linguistically, there is a lot of variation regarding the linguistic means of how clausal dependency or subordination is encoded (Cristofaro 2003).

In the following, some examples of hypothetical routes for purported developments of hypotactic from paratactic clauses will be presented in Indo-European languages and beyond, thereby highlighting some paradigmatic scenarios in the evolution of object clauses, relative clauses, and adverbial clauses. We will limit the discussion to the evolution of finite subordinate clauses.

2. Developmental routes in complement clauses

2.1. Evolution of *that*-type complementizers from paratactic structures with cataphoric demonstratives?

In the modern English and German sentences in (1a) and (1b), the object/complement clause is introduced by the complementizer *that* or *dass*, respectively.

(1) a. Copernicus believes strongly **that** the sun is the center of the universe.

b.

Kopernikus	glaubt	fest,	**dass**	die	Sonne	das	Zentrum
Copernicus	believes	strongly	that	the.NOM	sun	the.NOM	center

des	Universums	ist.
the.GEN	universe.GEN	is

The West Germanic *that*-type complementizer is homonymous with the 3. ps. neuter form of the demonstrative pronoun in the accusative or nominative case. This has given rise to the diachronic hypothesis that it goes back to a cataphoric use of the demonstrative in a combination of two loosely combined clauses where it was the complement of a *verbum dicendi*, *sentiendi,* or *cogitandi/putandi* and pointed to the content of the second clause. Scenarios have been put forward according to which the origin of the development is seen in a fully paratactic sequence of two main clauses as in the following constructed examples from modern Faroese: *eg sigi tað: hann kemur* 'I say that: he comes.' > *eg sigi at hann kemur* 'I say that he comes.' (cf. Lockwood 1968, 223). This strong hypothesis has been prominent in functionalist usage-based approaches to syntactic evolution. It has been seen as a hallmark case of syntacticization, defined by Talmy Givón as "a process by which flat, paratactic discourse pragmatic structures transform over time into tight, hierarchic syntactic structures" (Givón 1979, 82f.). It has also been taken up in neighboring disciplines, e.g. by the cognitive psychologist Michael Tomasello, who in a book on the evolution of language, summarized the origin of object clauses as follows: "If someone expresses the belief that Mary will wed John, another person might respond with an assent *I believe that*, followed by a repetition of the expressed belief that *Mary will wed John* – which become telescoped into the single statement *I believe that Mary will wed John* (a sentential complement construction)" (Tomasello 2010, 302).
Many diachronic accounts of this type have focused on the evolution of Germanic object clauses introduced by the THAT-type complementizer based on the neuter form of the demonstrative pronoun (cf. also Hopper & Traugott 2003 on English). However, if one looks beyond West Germanic, it seems that the role of the THAT-type complementizer has been overestimated. The Nordic *at*-complementizer is probably not derived from the demonstrative but from the local preposition *at* via an intermediate step that persisted into Old Norse, where *at* was an infinitival complementizer (Faarlund 2004, 2015). In East Germanic, there is also no evidence for the exact parallel type of THAT-complementizer that can be argued to be derived from a cataphoric demonstrative use. In Gothic, *ei* and

patei (pata-NOM/ACC + *ei*, cf. Miller 2019) are the two major complementizers, but *thata* alone is not attested as a complementizer, only the compound form with enclitic *-ei*, whose etymology is not entirely clear (it might be related to the IE. pronominal stem **e-*, **i-*, cf. Lehmann 1986, 99). A further potential problem is that such a diachronic scenario would involve several quite radical grammatical changes, both in sentential architecture, in the grammatical properties of THAT along with changes in grammatical person related to indexical shift (cf. *Copernicus states that: **I** assume the sun to be the center of the universe* > *Copernicus states that **he** assumes the sun to be the center of the universe*) (Axel-Tober 2017).

Outside of Germanic, there is hardly any evidence for full grammaticalization along the path demonstrative > complementizer (but see Kuteva et al. 2019, 155 for some potential examples). For Slavic, a parallel scenario has been proposed in some traditional work, but upon closer inspection, a development from an agreeing relative via a relative particle/complementizer is empirically more plausible (Meyer 2017): Complementizers in Slavic complement clauses are mostly cognates of relative particles rather than of demonstratives. For example, Rus. *čto* is ambiguous between a declarative complementizer, a *what*-type-interrogative/relative pronoun (used in free relatives), and a *wh*-based relative particle in colloquial language, whereas *(è)to* 'that-demonstrative' is morphologically unrelated. In Old Church Slavonic, *iže* (and variants thereof) is sometimes used as a declarative complementizer or occasionally as a non-agreeing relative marker.

Besides Germanic, Romance is a further language family for which there is extensive literature on complementizers of pronominal origin. Unlike in Germanic, the declarative complementizers are not based on the demonstrative stem, but on the relative pronoun, which is of interrogative origin (< IE. **k^{w}i/*k^{w}o-*). Most Romance complementizers ultimately originate in Latin interrogatives. Though the declarative complementizers mostly come from Latin *quia* 'because, that' (e.g., Italian *che*, French *que*, Spanish *que*) (REWOnline 6954) or Romanian *că* from *quod* 'that, because' (REWOnline 6970), these Latin complementizers are themselves interrogative-based. Note that *quia* is actually the older form of the accusative neuter plural of the pronoun stem *qui* (Walde & Hofmann 1954, 405), and *quod* is the nominative/accusative neuter singular form of the relative pronoun (Walde & Hofmann 1954, 405).

In sum, Germanic, Romance, and Slavic seem to be united in deriving their major declarative complementizers from relativizers. This, in turn, suggests that complement clauses have not developed from discourse juxtaposition, but from relative structures. In the next section, we will discuss the evolution from explicative or correlative relative clauses.

2.2. Complement clauses originating in explicative or correlative relative clauses

2.2.1. Indo-European

In general, in the realm of finite hypotaxis/subordination, relative clauses seem to be much older than complement clauses in Indo-European. Many older Indo-European languages do not primarily show complement clauses in the form of finite clauses introduced by complementizers, but rather infinite small-clause structures, (*accusativus cum participio/adjectivo constructions*) and constructions with abstract nouns or pronouns modified by relative clauses that explicate their content (Lühr 2004, Viti 2015). In Hittite and Old Indian, such clauses already showed uninflected complementizers (*kuit* and *yád*, (2)).

(2)

gr̥ṇé	tád	indra	te	**śáva**	
sing.IND.PR.MID1SG	this.ACC.N.SG	Indra.VOC	your	strength.N-ACC.SG	
upamáṃ	devátātaye	/ **yád**	dháṃsi		vr̥trám
supreme.ACC.N.SG	divinity.F-DAT.SG	that	kill.IND.PR2SG		Vr̥tra.ACC
ójasā					
force.N.INSTR.SG					

'I sing this supreme strength of yours, O Indra, for the divinity, that you kill Vr̥tra with your force.'
(Vedic; Viti 2007, 215)

It is thus most likely that in Indo-European, finite complement clauses did not evolve from paratactic structures, but from relative clauses explicating the content of overt or silent abstract nouns (Lühr 2004, Viti 2007, 2015, also Hettrich 1988, 407). Delbrück (1893, 1897, 1900) referred to what is called 'complement clauses' in modern linguistics as explicative relative clauses. In more recent approaches, such relative clauses are analyzed as explicating the content of light nouns (like *something*), content nouns (like *rumor*), or pronouns/pronominal adverbs. Axel-Tober (2017) shows reflexes of this in Old High German texts, notably in Otfrid where *thaz* is used as a relative complementizer (very much like English *that*) in ordinary relative clauses as well as in clauses associated with a noun phrase (like 'the story', 'the thing') or a pronominal correlative element (*thaz* 'that') in the context of *verba dicendi*, *sentiendi*, *putandi/cogitandi* etc.:

(3)

Ni	dróstet	iuih	**in thiu**	**thíng**,	\ **thaz**	íagilih	ist
NEG	console.2.PL	you.2.Pl	in the.INSTR	thing	that	everybody	is
édiling,							
nobleman							

'Do not take comfort in the fact that everybody is of noble descent'
(Old High German, Otfrid; adapted from Axel-Tober 2017, e40)

Apparently, the equivalent construction is still productive in Bengali, cf. (4). Bayer (2001) proposes that the clausal complementation structure arose from the noun-modifying construction via a reanalysis that involved the deletion of a "dummy NP" (like *story*) and a recategorization of the relative operator *je* in SpecCP as a complementizer in C^0. Axel-Tober (2017) proposes a similar reanalysis in the evolution of the *thaz*-complementizer in pre-OHG times.

(4)

chele-Ta	**e**	**kOtha**	jane	na	**je**	baba	aS	-be
boy-CL	this	story	knows	not	COMP	father	come-will	

'The boy does not know it that his father will come.'
(Bengali, adapted from Bayer 2001, 21)

Further cross-linguistic comparison suggests that the explicative relative structure might in turn have developed from an inverse correlative structure. Correlative relativization was a typical feature of the older Indo-European languages (Haudry 1973, Lehmann 1984) and also a Proto-Germanic phenomenon both in the nominal and adverbial domain (Kiparsky 1995, Axel-Tober 2007, 2012, 2023 in historical German), cf. also section 3 below.

In synchronic theoretical approaches, the classical view that so-called object clauses are syntactically complements and semantically propositional arguments of their matrix predicates has been challenged recently. Analyses have been advocated that treat such clauses also synchronically as relative clauses (Arsenijević 2009, Kayne 2014) and semantically as predicates (Kratzer 2006). Moltmann (2020) specifically proposes that apparent sentential complementation structures involve an underlying attitudinal object selected by the verb whose content is provided by the embedded clause. If such analyses are on the right track, they would cast new light on the putative diachronic processes, which would then potentially involve a form of continuation of the original 'explicative' relative structure.

2.2.2. Ugric

As was shown in the last section, evidence from Germanic, Bengali, Old Indic, and arguably from Romance and Slavic points to a hypothetical route from relative clauses in explicative or correlative structures to declarative complement clauses. Specifically, a relative-like structure where the subordinate clause explicated the content of an abstract and potentially silent nominal element seemed to have played a crucial role. Outside Indo-European, evidence from Old Hungarian and its conservative sisters Khanty and Mansi provide a window into this pathway. É. Kiss (2023) argues that the object clauses in the form of a finite CP-complement with an uninflected complementizer as a head in C^0 arose via reanalysis of an inverted correlative structure. In Hungarian, the declarative complementizer is *hogy*, which etymologically goes back to a manner pro-adverb 'how'. Old Hungarian witnesses a (reversed) manner correlative construction with the relative pro-adverb *hogy* 'how' and the demonstratives correlatives *úgy*, *így* 'so'. In the contexts of verbs of communication, the correlative structure, i.e., an adjunction structure was reanalyzed as a truly hypotactic structure, i.e. a clausal complementation structure: the 'adjunct clause' developed into an 'argument clause'. The evidence from Old Hungarian thus points to a hypothetical route in (5), leading from a weakly hypotactic structure, the correlative construction, to a truly hypotactic structure, where the subordinate CP is a constituent of the matrix CP, namely a complement of V eliciting agreement on V.

(5) correlative clause > explicative clause > complement clause

Sentences as in (6) represent the intermediate step in this development. They are only formally identical to correlatives in their external syntax. Interpretatively the subordinate clause explicates the pro-adverb *úgy* 'so'.

(6)	vǵ	mond	zenth	Gergel	doctor,	**hoǵ**	az	ǫrdǫg	ez	fǫld-et
	so	saysS	Saint	Gregory	doctor	as	the	devil	this	earth-ACC
	kereng-i.									
	circle-OBJ.3SG									

'So says Doctor St Gregory as/that the devil is circling this earth.'

(Hungarian; É. Kiss 2023, 118)

The evolutional paths of the Hungarian *hogy* and the Germanic *that*-type complementizers as described in Axel-Tober (2017) are thus strikingly similar. While the Germanic complementizers started out as the relative pronominal correlate of a demonstrative object, *hogy* was originally the correlate of a demonstrative manner adverb.

Summarizing so far: Can the development of complement clauses from relative clauses be argued to involve a change from parataxis to hypotaxis? *Stricto sensu*, this is not the case as relative clauses are themselves semantically and syntactically dependent clauses. Even though relative clauses of the correlative type are not embedded into the main clause, they are treated as dependent clauses in most approaches, i.e., as adjoined at the left or right peripheries of their main clauses (cf. section 3.1. for further discussion). Under a gradient view of hypotaxis (cf. Lehmann 1988 for an approach), it could be argued that we are dealing with the development of a weaker hypotactic into a stronger hypotactic structure.

2.3. Development of declarative complement clauses from interrogatives?

Along with *d*-pronouns, *wh*-pronouns form a common source of declarative complementizers. Within Indo-European languages, interrogative-based declarative complementizers occur, for example, in Slavic or Romance languages (cf. section 2.1.), but besides them also in Georgian, Mandarin Chinese, or in !Xun, a Khoisan language (see Heine & Kuteva 2007, 242ff.). Heine & Kuteva (2007, 243) discuss a hypothetical route from a paratactic source structure as in (7a) where the second sentence is an independent *wh*-question. In the first step, the development results in a structure like (7b) "where the question word assumes the function of a complementizing pronoun and the interrogative clause [turns] into a complement clause" (Heine & Kuteva 2007, 243). According to the examples that Heine & Kuteva (2007, 243) give to exemplify this development (see (8a, b)), the independent question develops into an embedded *wh*-question (and nothing actually happens to the *wh*-pronoun yet, since in (8b), it simply continues to be a *wh*-pronoun and nothing else). In some languages, however, the *wh*-pronoun further develops into a declarative complementizer corresponding to English *that*. This developmental path would correspond to the parataxis-to-hypotaxis hypothesis.

(7) a. [S1] [QW + S2?]
    b. [S1 [CPL + S2]]

(8) a. [I don't know:] [What does he want?]
    b. [I don't know [what he wants]]

However, there are serious doubts about this explanation. First, as Heine & Kuteva themselves admit, so "far, no historical data have been found to corroborate this reconstruction" (Heine & Kuteva 2007, 243, fn. 16) of the development from (7a) to (7b). It seems much more plausible that complementizers evolve from *wh*-pronouns in a structure where the interrogative clause is already embedded. This would, of course, no longer be in line with the parataxis-to-hypotaxis hypothesis. Second, there is another way that complementizers can arise from *wh*-pronouns. As with *d*-pronouns (see above), the most obvious way is via relative clauses. Under such a scenario, *wh*-relative pronouns first develop into relative complementizers and are then generalized as declarative complementizers. Weiß (2020) reconstructs this development for Yiddish *voś*, which first replaced the original relative pronouns *der/di'/doś* 'that, which' and then developed into a complementizer introducing relative clauses. In a further step, it could also be used as a declarative complementizer:

(9)	Veyś=tu	den	nit	**voz**	unz	Ari	cu	gihert?
	Know.2SG=you	PRT	NEG	what	us.DAT	Ari	to	belongs

'Don't you know that Ari belongs to us?'
(Yiddish; Weiß 2020, 43)

To conclude, the development of declarative complementizers from *wh*-pronouns does not provide evidence for the parataxis-to-hypotaxis hypothesis. The two most probable paths of origin start with relative clauses and embedded *wh*-questions respectively, i.e., with hypotactic structures.

2.4. Further complementizer sources and hypothetical routes

Besides pronominal origins (of the *d*- or *wh*-type), which may involve relative structures as argued in the last sections, cross-linguistic surveys on the origins of declarative complementizers mention, among others, adverbials (notably locatives), light nouns such as 'thing' or 'matter' and *say*-type verbs as important lexical sources (see Boye et al. eds. 2016, Chappell 2008, Heine & Kuteva 2007). As Chapell (2008, 3, fn. 3) concedes, "some of the sources could turn out to be related by one and the same grammaticalization pathway". In the following, we will show that this is indeed the case and that many developments are based on relative structures.

2.4.1. Source constructions with adverbs

The Greek complementizer *pu* that is used in the finite complements of some verb classes etymologically goes back to a local adverb, but there is no direct development from parataxis to hypotaxis. Rather, *pu* first grammaticalizes into a relative adverb and later into a general relativizer (Roberts & Roussou 2003, 120). In a further step, it apparently develops into a causal subordinator before it comes to be used as a declarative complementizer. According to Nicholas (1998), the declarative *pu*-complementizer presumably resulted from a reanalysis of causal *pu* (cf. the English paraphrases *Timosa was angry because you left* > *Timosa was angry that you left*). Outside Indo-European, similar developments have been reported for Semitic. Proto-Semitic did not have finite complement clauses. Deutscher (2000) reconstructs the development of finite complement clauses in Akkadian with the complementizer *kīma* based on evidence from Old Babylonian. He argues that they developed from causal adverbial clauses in bridging contexts with speech-related verbs where both a causal and a factive interpretation were possible, see the first attested example in (10). *Kīma* eventually goes back to a comparative preposition but has developed adverbial interpretations (temporal, causal) later.

(10)	PN	šupur-ma	**kīma**	nār-um	sekrat	šudi-šum.
	PN	write.IMP-P	*kīma*	river.NOM	blocked.STATIVE(FSG)	make known.IMP-to him

‘Write to PN and inform him because/that the river is blocked.’
(Old Babylonian, Whiting 1987, 40: 3; cited from Deutscher 2000, 43)

Structurally, the change thus involves a development from “peripheral elements to arguments” (Deutscher 2000, 48) as adverbial clauses have the status of adjuncts. If at all, in these Akkadian and Greek pathways we are again dealing with developments from weaker to stronger hypotaxis.
Biblical Hebrew also provides a window into pathways of how complement clauses developed. As Givón (1974, 1991) argues, here the relativizer *’asher/she* could have been “transmitted” into V-Comp paradigms after accusative-governing utterance and cognition verbs that take accusative nominal complements. The examples that he cites, notably those with dummy and pronominal objects, cf. (11), show parallels to the explicative clauses discussed in the literature on the old Indo-European languages.

(11)	shamd'nu	**'et-'asher**	hovish	YHWH	'et-mey	yam	Suf.
	Hear.PERF.1PL	ACC-SUB	dry.PERF.3MS	YHWH	ACC-water-of	sea-of	Suf

‘We've heard that God had dried up the waters of the Red Sea.’
(Biblical Hebrew, Joshua, 2.10; adapted from Givón 1991, 290)

’asher/she is ultimately of locative origin, but it did not directly develop into a declarative complementizer, but first into a relativizer. An older declarative complementizer (*ki/vehineh*) is also attested in these environments. What we thus see in the transmission of Biblical Hebrew is not the origin of finite sentential complementation but its renewal according to the distinction made by Harris & Campbell (1995).

2.4.2. Source constructions with nouns

Besides pronouns and adverbs, nouns are another important source (Heine & Kuteva 2007, 230-236). In many languages, light nouns with a generic meaning developed into complementizers, that is, nouns meaning ‘person’, ‘thing’, or ‘matter’ are grammaticalized into complementizers introducing subject or object clauses, and nouns meaning ‘place’, ‘time’, ‘manner’, ‘kind’, ‘way’ into complementizers introducing adverbial clauses (see section 4 on adverbial clauses). Declarative complementizers (corresponding to English *that*) grammaticalized from a noun meaning ‘thing’ or ‘matter’ are attested in several African languages and in Japanese (see below). Note that in many cases mentioned in the literature, the noun together with a relative marker develops into a pronoun that then can head a free relative clause (Heine & Kuteva 2007, 230). This is the case, for instance, with the examples from Ewe (a Niger-Congo language), where the nouns meaning ‘person’ and ‘thing’ together with the relative marker developed into *amé-si* (person-REL) and *nú-si* (thing-REL) which correspond to English *who(m)* and *what* (Heine & Kuteva 2007, 231). We will neglect such cases in the following.

It is assumed that nouns develop into complementizers in two ways (Heine & Kuteva 2007). The first pathway is via relativization, that is, the noun in question is modified by a relative clause and develops then into a complementizer (and the relative clause is reanalyzed as a complement clause), see (12a–b) (Weiß 2020, 37):

(12) a. [CP … [N [RC …]]]
b. [CP … [CP Compl …]]

The E1 dialect of !Xun (Northern Khoisan; Namibia) (Heine & König 2015) provides an example: the noun *tcí* 'thing' together with the relative suffix *–à* developed into the complementizer *tcá*:

(13) a. ! *tcí* 'thing' + relative suffix *–à* -> *tcá*

b.

mí	tsà'á	**tcá**	hȁ	kȍh	gù	dshàú
1SG	hear	COMPL	N1	PAST	take.SG	wife

'I heard that he got married.'
(Xun, E1 dialect, Northern Khoisan; Heine & König 2015, 285)

Example (13b) represents the case where the original relative marker survived as part of the new complementizer. Heine & Kuteva (2007: 235) give another example from the W2 dialect of !Xun (*tcí* 'thing' and the relative clause marker -*è*, hence tcí-è > *tcé*) and one from Ik (a Nilo-Saharan language spoken in northeastern Uganda) (Heine & Kuteva 2007, 233f.). This last example is interesting for two reasons. First, even though the lexeme *tóimɛn* no longer exists as a noun and has "no function other than introducing complement clauses" (Heine & Kuteva 2007, 234), it is still obligatorily case-marked (for accusative or dative). Second, it was grammaticalized as a declarative complementizer without the relative marker *na*. Since there are other generic nouns in Ik that developed into a complementizer together with the relative marker, it is reasonable to assume the relativizing strategy for *tóimɛn* as well (for the details see Heine & Kuteva 2007, 233ff.).
However, in other cases, where the noun alone developed into a complementizer, Heine & Kuteva (2007) postulate a second pathway for the grammaticalization of nouns into complementizers. They assume a source structure where "the complement clause is added to the complement noun without any formal marking" (Heine & Kuteva 2007, 235), that is, with two independent clauses. They give two examples of such a parataxis-to-hypotaxis development. The first comes from Nama, a Namibian Khoisan language, where the noun *!xái-s* (*!xái-sà* oblique case) 'matter, story' developed directly into a complementizer. Just like *tóimɛn* in Ik, it is still inflected like a noun even though it is clearly a complementizer, see (14).

(14)

tiíta	ke	kè	ǀ'úú	'ií	!úũ-	ts	ta	**!xái-**	sà.
1.SG	TOP	PAST	NEG.know	PAST	go-	1.SG.M	IMPFV	COMP-	3.SG.M

'I didn't know that you were going.'
(Nama (Central Khoisan); Heine & Kuteva 2007, 235f.)

The second example is Japanese *koto* which also developed directly into a complementizer (Heine & Kuteva 2007, 236). However, both examples are not convincing for several reasons. First, the assumption of a parataxis-to-hypotaxis development in both cases does not take into account that relative clauses can be added asyndetically, that is, without overt marking. That seems to be the case with Japanese *koto*: Comrie & Horie (1995) analyze complement clauses introduced by *koto* as structurally comparable to relative clauses since both consist of a nominal head and a dependent clause, where the relation between both is not marked overtly. Therefore, the grammaticalization of *koto* need not be exceptional, but may well have involved relativization as well (Weiß 2020, 41). Second, the parataxis-to-hypotaxis proposal does not reflect the fact, either that there are clear cases of grammaticalization of the noun alone where we have enough historical evidence for a source structure with a relative clause. A relevant example of this kind is German *weil*, where the relative complementizer has not become part of the new complementizer (see section 4 on adverbial complementizers). The Nama example could be of this kind because we can observe exactly this type of development in other Khoisan languages (see the !Xun example in (13) above). Altogether, we can say that the grammaticalization of complementizers from nouns does not provide reliable evidence for the parataxis-to-hypotaxis hypothesis.

2.4.3. Source constructions with SAY-verbs

A further common source in the grammaticalization of complementizers is SAY verbs or *verba dicendi*. This has been reported for many languages in the African, South, and Southeast Asian regions and for Creoles (cf. the references in Chappell 2008 and Saito 2021, 17). For example, the Taiwanese complementizer *kong* in (15) is originally a speech verb.

(15) Ahui siong **kong** Asin m lai
Ahui think KONG Asin NEG come
'Ahui thought that Asin is not coming.'
(Taiwanese; Simpson & Wu 2002, cited from Saito 2021, 7)

In many cases, the development originated in a serial verb construction (Lord 1993 Klamer 2000, Roberts & Roussou 2003) of the type 'Ahui think say [CP …]'.
Diachronically, the reanalysis of the serial verb construction has been argued to have been driven by structural simplification through which the higher verbal head is eliminated and the morphologically and semantically impoverished *say*-item is reanalyzed as a C^0-element (Roberts & Roussou 2003, 126, Saito 2021). To the extent that the serial verb construction can be characterized as paratactic, "this change also signals a change from a paratactic to a subordinating construction" (Roberts & Roussou 2003: 126). However, in contrast to the traditional parataxis-to-hypotaxis scenario, it does not involve a union of two independent sentences.

3. Developmental routes in relative clauses

3.1. Correlative and post-nominal relative clauses

As already discussed in section 2, correlative clauses were an important relativization strategy in the old Indo-European languages and in ancient Ugric (cf. also Belyaev & Haug 2020 on their wide cross-linguistic distribution). Schematically, this type of relativization can be represented as follows:

(16) a. [Which **books** I recommend to him] Jack never reads **them**.
b. [$_{\text{S-matrix}}$ [$_{\text{RC}}$ (…) N …] [$_{\text{S-matrix}}$ … (Dem) …]
(adapted from de Vries 2002, 20)

The relative clause is not embedded into the matrix clause. In most analyses, it is externally adjoined, typically at the left periphery. The relative head is relative-clause internal. In the matrix clause, there is a resumptive pronoun, mostly a demonstrative. All old Indo-European languages show correlative relativization in varying forms, also called ‘correlative diptych’. It is the oldest form of subordination and can be reconstructed for the Proto language, even though it is not possible to reconstruct a relative pronoun or determiner due to the variation in forms in the ancient languages (Viti 2015, Ram-Prasad 2022). It is the standard form of relativization in Vedic and Hittite, where these clauses are also quite flexible regarding tense and mood, whereas it is less frequent in Latin and Ancient Greek. In the latter languages, embedded relative clauses show morphosyntactic restrictions (viz. *consecutio temporum*, *optativus obliquus*) (Viti 2015). Correlative relativization is also attested in Old Germanic, Slavic, and Baltic, but here other forms, notably post-nominal embedded relative clauses, prevail (Viti 2015, 462). The correlative diptych, with its external adjunction structure, has been characterized as a weak form of subordination/hypotaxis (Lehmann 1988, Kiparsky 1995). The more flexible tense and mood properties in Vedic and Hittite lend further support to this conviction (Viti 2015, 460). There is a controversy in the literature regarding the question of whether the relative clause in correlative relativization is base-generated in its peripheral adjunct position to IP or CP or whether it has moved from a modifier position adjacent to the correlate in the main clause. In the base-generation analyses, there are asymmetric and symmetric adjunction analyses (cf. Motter 2023 for a recent overview of the different approaches). Analyses with asymmetric adjunction are clearly not paratactic, they involve a syntactic relation between a dependent clause and a main clause, even though this relation is arguably not as hypotactic as in structures where the dependent clause is syntactically embedded into the main clause. Recently, Motter (2023) provides the argument that Hittite correlatives are not in a syntactically integrated position, not even as external adjuncts. The two clauses do not stand in a syntactic relation but are only connected at the level of discourse grammar. It has to be seen whether this type of analysis would also be appropriate for correlative structures in further old Indo-European languages. In Germanic, the two clauses in correlative structures differ in their internal syntax already in their earliest attestations. The relative clause e.g. typically shows word-order properties of dependent clauses (verb-final order).
Overall, there is a diachronic tendency in Indo-European through which embedded relatives have gained ground and correlatives have been marginalized, even though there are residues in many modern languages and varieties. It could thus be argued that there is a pan-Indo-European diachronic strengthening of hypotaxis in the realm of relativization. From this observation alone one can, of course, not conclude that

embedded forms of relative clauses have structurally evolved from correlative source structures. However, Bianchi (1999) argues that is probably no coincidence that correlative and post-nominal relatives in Latin (cf. (17a) and (17b)), Old English, and Hindi exploit the same relative morphemes. This fact can be seen as evidence for a structural commonality.

(17) a. [$_{CP}$ [$_{DP}$ **Quibus** [$_{NP}$ **diebus**]]$_i$ Cumae liberatae sunt obsidione],
which.ABL days.ABL Cuma released are siege.ABL
[[**isdem** **diebus**]$_i$ Ti. Sempronius prospere pugnat]
the same.ABL days.ABL Ti. Sempronius successfully fights
'T.S. wins a victory in the days in which Cuma is released from the siege.'
(Latin; Bianchi 1999, 86, based on Haudry 1973, glosses adapted)

b. ex [$_{DP}$ iis [$_{NP}$ [$_{NP}$ rebus] [$_{CP}$ **quas** gerebam]]] intellegebatis
from these.ABL things.ABL which.ACC did.1Sg understood.2Pl
'From the things I did you understood...'
(Latin ; Bianchi 1999, 87 ; glosses adapted)

According to the standard generative analysis, the relative morpheme is an independent anaphoric pronoun in post-nominal relatives and a relative determiner selecting the head of the antecedent in correlatives, which are head-internal. According to the raising analysis (Kayne 1994), the relative morpheme is a determiner selecting the relative NP-head in both types of relativization strategies. In both cases, the relative head originates within the relative clause itself. In the post-nominal relative, this is obscured by the fact that the 'head' (= *rebus* in (17b)) raises to the left of the relative determiner in order to enter into a checking relation with the external determiner (= *iis*). Under the raising analysis, it is thus not surprising that the two structures share the same relative morpheme and it is straightforward to postulate a diachronic transition between the two structures (Bianchi 1999).

3.2. From demonstrative to relative pronoun?

In the Indo-European languages, the relative pronouns derive from different sources: from **so-/to-* in Germanic, from **yo-* in Indo-Iranian, Ancient Greek, Armenian, and Balto-Slavic and from **k^{w}i/*k^{w}o-* in e.g. Hittite, Latin, Baltic, and Classical Armenic. Some relative uses might have developed secondarily from interrogative or indefinite uses of descendants of **k^{w}i/*k^{w}o-*, but in Italic, Hittite, and Tocharian, **k^{w}i/*k^{w}o-* is already found in the earliest stages (cf. Probert 2015, chapter 3, Ram-Prasad 2022, 57 for details).
The demonstrative-based relative pronouns in Germanic (*d*-relatives) have figured prominently in the Neogrammarian conceptualization of parataxis-to-hypotaxis developments. The West Germanic languages show *d*-relative pronouns from their earliest attestations. Furthermore, *d*-pronouns are attested in Gothic – in combination with a relative complementizer (Harbert 2007, 436). Outside Indo-European, demonstrative-based relatives can be found in Arabic and Chinese (see van Gelderen 2009 for further discussion).

3.2.1. Development from a paratactic structure with an anaphoric pronoun

Delbrück (1900: 445) proposes that relative constructions ultimately go back to a paratactic sequence in which an anaphoric pronoun referred back to a nominal in the preceding clause: "If the content of the second clause with the initial pronoun was such that the clause could be interpreted as a more or less insignificant addition to the first one, it became a subordinate clause and the pronoun a relative" [our translation]. Behaghel (1928, 766) uses the following constructed example to illustrate how the relative pronoun might have evolved from the anaphoric demonstrative:

(18)

das	ist	das	Lamm,	**das**	trägt	der	Welt	Sünde	>
that	is	the.NOM	lamb	that-DEM	carries	the.GEN	world	sin	

'this, is the lamb that carries the world's sin.'

das	ist	das	Lamm,	**das**	der	Welt	Sünde	trägt	
that	is	the.NOM	lamb	which-REL	the.GEN	world	sin	carries	

'this, is the lamb which carries the world's sin.'

From the perspective of modern linguistic theory, the downgrading of an independent clause would involve several radical syntactic (and semantic) reanalyses in one swoop – regardless of the question of which specific implementation of relative-clause analysis one adopts. It should be noted that again, we are dealing with a completely hypothetical route. Heine & Kuteva (2007, 226) schematize the purported development as follows:

(19) [S1 + S2] juxtaposition to S1 [S2] relativization

They provide an example from Old Norse as evidence for how this alleged development might have happened. In example (20) from a runic inscription, the second sentence contains the demonstrative pronoun *sā* that refers back to the proper noun *Eyvind* in the first sentence. Now, Heine & Kuteva (2007, 226), following Zeevaert (2006), assume that it is ambiguous "between the (paratactic) demonstrative and the (hypotactic) use of *sā* as a non-restrictive relativizer". Note that Zeevaert (2006, 21), who first introduced this evidence into the discussion, translates *sā* with a personal pronoun in German. However, it would have been more appropriate to use the *d*-pronoun *der* 'this', which in German occurs frequently in anaphorical use (see Weiß 2020 for more details).

(20)

stikuR	karþi	kubl	þau	aft	auint
Stig.NOM.SG	made	monuments.ACC.PL	these.ACC.PL	after	Eyvind

sunu	sin	**sa**	fial	austr.
son.ACC.SG	his	he	fell	east.

i. 'Stig made these monuments after his son Eyvind. He died in the east.' or
ii. 'Stig made these monuments after his son Eyvind, who died in the east.'
(Old Norse (ca. 800 AD); Zeevaert 2006, 21; glosses added)

This ambiguity triggers the reanalysis of the demonstrative pronoun as a relative pronoun and the degrading of the independent sentence to an embedded clause. However, there are several shortcomings in Heine & Kuteva's (2007) argumentation (Weiß 2021; 45f.). First of all, there is little evidence for the purported ambiguity of (20) and thus for “the transition from paratactic to hypotactic forms of clause combining” (Heine & Kuteva 2007, 225). It is not clear why ambiguity should arise in (20). Note that in Old Norse, relative clauses were introduced by the complementizers *er* or *sem*, whereas the demonstrative pronoun *sā* (if at all) only occurred as (part of) the antecedent (Faarlund 2004, 264). It is thus most likely that *sā* in (20) is not a relative pronoun, but a demonstrative pronoun that refers anaphorically back to the proper noun *Eyvind* in the preceding sentence.

Zeevaert (2006, 20) discusses a further example from Old Norse (Stone of Stentoften, KJ96 DR357; Bl3, see http://www.runenprojekt.uni-kiel.de/abfragen/default.htm) where it is not unlikely that *sā* is a relative pronoun that introduces a relative clause without complementizer, cf. (21).

(21)	herAmala[u]saR	(A)rg(j)u	wē(la)d[a]ud[e]	**sā**	þat	bAriutiþ.
	defense-less	with effeminacy	treacherous	who	that	breaks

‘Perplexed by evil, anyone who destroys this will die a treacherous death.’
(Old Norse (ca. 800 AD); Zeevaert 2006, 21; glosses added)

However, if anything, it is a free relative clause introduced by a demonstrative pronoun – as was common in the earliest documented Germanic languages such as Gothic, Old English, or Old High German (Harbert 2007, 269f.; Weiß 2016; Axel-Tober 2012). Wagener (2017) mentions a slightly later runic inscription, which presents a parallel version of this text. Interestingly, in this inscription, *sā* is replaced by *sAR* which seems to be a form contracted of *sā* and the complementizer *er*. According to Wagener (2017), *sā* never developed into a relative pronoun but belongs to the nominal shell that embeds the relative clause.

3.2.2. Development from a paratactic structure with a cataphoric demonstrative

In a different type of scenario, the *d*-relative originates from a structure with an asyndetic relative clause, i.e., a relative clause that is not introduced by an (overt) relative pronoun or relative complementizer.

(22)	in	droume	si	in	zelitun	then	weg	sie	faran	scoltun
	in	dream.DAT	they	them.DAT	told	the.ACC	way	they	go	should

‘They told them in their dream the way they should go.’
(Old High German; Otfrid, adapted from Dal ^{3}1966, Axel-Tober 2012, 226)

According to a widely proposed hypothetical route, the *d*-relative goes back to a structure with a cataphoric demonstrative at the end of the first clause, followed by an asyndetic relative clause. The demonstrative developed into a relative pronoun via a shift of the clause boundary (Dal ^{3}1966: 198, Helgander 1971, 136–181, Heine & Kuteva 2007, 225ff.). This purported development shows parallels to the traditional clause-boundary shift scenario of the *that*-type complementizer and has been challenged on similar conceptual grounds (Lenerz 1984, Axel-Tober 2012, 2017). It is also empirically not well supported as asyndetic relative clauses are only attested in Old High German and are absent from Old English and Gothic (Harbert 2007, 436). Be that as it may, even if this development had more plausibility, it would not be an example of a parataxis-to-hypotaxis development. The asyndetic clause in the source construction was arguably already a subordinate clause, not only interpretatively, but also syntactically. In modern syntactic theory, it would probably be analyzed as containing a zero complementizer or the like (cf. Pittner 1995 for an implementation).

3.2.3. Renewal of relativizers in pre-existing hypotactic sources

We can speculate with reason that West Germanic *that*-type complementizers may have evolved principally into relative and declarative complementizers in two different ways. First, as described above in section 3.1., from a correlative source structure where *d*-pronouns could also occur (Axel-Tober 2012). The second pathway involves rebracketing, that is, a demonstrative pronoun that heads a relative clause first develops into a (neuter) relative pronoun, which then further develops into a relative complementizer and eventually into a declarative complementizer (Weiß 2019, 2020, 2021). Note that this pathway also involves a shift of a clause boundary, not of a boundary between independent sentences, but of a boundary that separates an embedded clause from the main clause. This development can be illustrated by the example of the Gothic relative complementizer *þatei* 'that'. Weiß (2020), based on Delbrück (1909), proposes that it developed as sketched in (22a–d). In the source structure in (23a), the object in the main clause is a demonstrative pronoun that takes an explicative relative clause introduced by the complementizer *ei*, which specifies the content of the demonstrative pronoun (Axel-Tober 2017: e42). As can be seen in (23b), the demonstrative pronoun first develops into a relative pronoun and changes its syntactic position from outside the relative clause to one within it (this change is an instance of rebracketing in the sense of Weiß 2019, 2021). In a further step, the relative pronoun univerbates with the relative clause complementizer (23c), thus forming the new relative complementizer *þatei*, which finally develops even into a general subordinator that can introduce complement clauses as well (23d).

(23) a. [$_{CP}$ … [$_{DP}$ þata [$_{CP}$ [$_{C°}$ ei] …]]]
b. [$_{CP}$ … [$_{DP}$ [$_{CP}$ þata [$_{C°}$ ei] …]]]
c. [$_{CP}$ … [$_{DP}$ [$_{CP}$ [$_{C°}$ þatei] …]]]
d. [$_{CP}$ … [$_{CP}$ [$_{C°}$ þatei] …]]]

This developmental path is not restricted to Gothic. Poletto & Sanfelici (2018) report a very interesting recent development in Marebbano (a Rhaetoromance verb-second variety) where a demonstrative pronoun occurs in prepositional relative clauses (e.g., *de chël che* 'of that that' or *a chell che* 'to that that'). They analyze the demonstrative as part of the relative clause, that is, the development has reached a level that is comparable to (23b) or even (23c) in the Gothic case (see also Weiß 2020).

For *that*-type complementizers in the West Germanic languages – i.e., Germ. *dass*, Dutch *dat*, and Engl. *that* – we can thus assume two possible ways how they emerged. First, it is possible that they developed structurally in the same way as Gothic *þatei*. The only difference is that the original relative-clause complementizer – of the form *þe/ðe/the/de* – vanished with the result that only the pronoun remained and was eventually reanalyzed as a complementizer. This analysis was proposed by Weiß (2020). However, it is also possible that *that*-type complementizers emerged from correlative structures as has been proposed by Axel-Tober (2012). In both scenarios, we are not really dealing with innovation of relative hypotaxis, but rather with a form of renewal. The source constructions were already relative structures with uninflected relative complementizers (like modern English *that*), which are probably at least as old, or even of older origin than pronominal *d*-relativizers. According to Harbert (2007, 434), the "template" '*d*-relative pronoun+complementizer' (as in Gothic *þat(a)+ei*, Old English *se+ þe*, Old High German *ther + the /therde*) must have been inherited from Proto-Germanic.

4. Developmental routes in adverbial clauses

4.1. Adverbial clauses with origins in (correlative) relative clauses

Cross-linguistically it can be observed that languages differ with respect to the question of which clausal relations they may convey by means of an adverbial subordinate clause or by constructions with coordination and juxtaposition. Many languages have non-anaphoric adverbs that express temporal, local, or manner relations alongside the respective types of clausal adverbials. Interestingly, such "primary adverbial clauses" (Viti 2013) can be paraphrased by a combination of a semantically relatively empty head noun and a relative clause, cf. *We'll go at the time at which Tom gets here* (temporal), *I'll meet you at the place at which the statue used to be* (locative), *She spoke in the way in which he had taught her to* (manner) (cited from Thompson, Longacre & Hwang 2009 ex. 18). It is thus not surprising that in many cases primary adverbial clauses also originate in relative constructions diachronically. Relative adverbs or pronouns are major cross-linguistic sources of adverbial subordinators (Kortmann 1997, Heine & Kuteva 2007, 250f.). In the old Indo-European languages, primary adverbial clauses often show a tight association with relative clauses. The correlative diptych is the prevailing strategy for adverbial subordination in Hittite and Vedic (Haudry 1973; Viti 2015, 319; Kiparsky 1995), it is attested in all old Indo-European languages. This type is still very productive in Lithuanian (Ambraszas 1997) and, beyond Indo-European, in Hungarian (Bhatt & Lipták 2009). Since correlatives are often formally identical to left-dislocated embedded relatives, Axel-Tober (2023) argues that the old Germanic adverbial correlatives might have been reanalyzed as embedded adverbial clauses. For example, the German causal complementizer *da* derives from the temporal adverb. The source construction was most likely not a juxtaposition of independent clauses, but a construction where *da* (> Old High German *thō*) was a relative adverb. In the following example from the Old High German Tatian translation, it is combined with the relative complementizer *the* (cf. 3.2. above on the combination of *d*-pronouns + complementizers in Germanic); the matrix clause shows a correlative use of the demonstrative adverb as part of a correlative diptych. After *thō/d*a had been reanalyzed as a complementizer (a C^0 element), it was in use as a general temporal complementizer for some centuries until it underwent the wide-spread meaning change from temporal to causal interpretation (Axel-Tober 2012).

(24)	[$_{\text{CPrelative clause}}$	**thothe**	*e*rstigun	sine	bruoder	[$_{\text{CPmatrix}}$ **tho**	ersteig	her	úf.]
		then.COMP	ascended	his	brothers	then	ascended	he	up

'when his brothers had gone up, then he also went up.'
(Old High German; Tatian, adapted from Axel 2007, 233)

It has often been noted that many adverbial subordinators in the old Indo-European languages derive from relative stems. In her survey, Viti (2015, 327) summarizes that the Hittite subordinators *kuit* 'because, that', *kuitman* 'when, while, as long as, as, until', *kuwapi* 'at the time when, then when' go back to the Proto-Indo European relative and interrogative pronoun *k^{w}i/k^{w}o-*. She also mentions that the Lithuanian relative pronoun *ka-* is the source of a wide range of adverbial subordinators with different temporal, causal, and conditional meanings (*kad*, *kai*, *kada, kol*, *kadangi*, *kad ir*, *kaip*). The Vedic relative stem *ya-* is the basis for the adverbial subordinators *yadā́, yádi, yáthā, yátra, yátas, yā́d* etc. The same is true for Ancient Greek *ὡς* 'how, as, after that, because, so that, that' and *ὥστε* 'so that, under the condition that, in order to' (< **yo-*) (ibid.).
In the case of adverbial subordinators of relative origin, we are not dealing with a route from fully-fledged parataxis-to-hypotaxis as relative clauses are subordinate structures themselves. It should be noted, however, that correlative relatives are analyzed as less hypotactic than other types of relative or adverbial clauses in some frameworks or even as fully syntactically paratactic (see Motter 2023 on Hittite).

4.2. Adverbial clauses with subordinators based on adverbs, nouns, or prepositions

4.2.1. Source constructions with adverbs

Heine & Kuteva (2007, 250f.) present one case where they assume an adverbial source in a bi-sentential source structure. They argue that an adverb meaning, e.g., 'here' at the beginning of a sentence can be reanalyzed as a complementizer (see 24). This again would be in line with the traditional parataxis-to-hypotaxis scenario that was proposed, for example, by Dal (1966, 187).

(25) From adverb to clause subordinator (SUB)
  a. [main clause] – [adverb + main clause]
  b. [main clause] – [[SUB + subordinate clause]]

Heine & Kuteva also provide relevant examples from Lingala, a Bantu Language, and Albanian (26a, b) where the adverb meaning 'here' developed into a temporal and a causal complementizer, respectively:

(26) a. **áwa** oy´ɔ olingí tɛ, tokotínda mwána mosúsu
  here you you.come NEG we.FUT.look child other
  'Since you don't come, we'll look for another boy.'
  (Lingala; Heine & Kuteva 2007, 251)
  b. **ke** s'fole ti, [. . .].
  here not PTCPL.say.2.SG
  'Because you did not say anything, [. . .].'
  (Albanian; Heine & Kuteva 2007, 251)

4.2.2. Source constructions with nouns

Further common sources of adverbial subordinators are nouns and prepositions. Here again, we can often hypothesize routes through which adverbial complementizers emerge via relativization. Remember that light nouns meaning ‘thing’ or ‘matter’ could develop into declarative complementizers in a source structure where they take an explicative relative clause and they then grammaticalize (with or without the relative marker) into a complementizer. The same can be observed with light nouns meaning ‘time’ or ‘place’, for instance. Kuteva et al. (2019, 438) give examples for the noun meaning ‘time’ developing in a temporal complementizer from languages as diverse as Japanese (*toki*), Chinese (*shi*), or Kupto (a West Chadic language of Nigeria) (*sàrtí*), to which could be added English (*while*) and German (*weil*), which both evolved from a noun with the meaning ‘while, time span’ (Weiß 2019, 2020). Note that German *weil* first developed as a temporal complementizer that only later acquired a causal meaning (Eberhardt & Axel-Tober 2023, Weiß 2019, 2020, 2021). Viti (2013; 2015, 322) reports evidence from Old Irish, where these adverbial relative clauses may be built based on nouns denoting ‘day’ or ‘manner’, from Hittite temporal clauses that consist of a relative construction with the head nouns denoting ‘time’, ‘day’ or ‘night’ and from Vedic temporal clauses with the general relative subordinator *yád*.

The German complementizer *weil* ‘because’, which grammaticalized from the noun *Weile* ‘while, time span’, is an interesting case since it apparently allows us to reconstruct its emergence in two ways. The first possibility is to explain its emergence on the basis of an internally-headed relative clause in a correlative structure like in the Middle High German example in (27), where the noun phrase *die wîl* (an accusative DP with an adverbial reading as if it were the object of a preposition) is in SpecCP followed by the relative complementizer *daz* in C^0. (In (27) *daz* could also be the neuter definite article in the NP *daz fiwer*, under this analysis C^0 would host a silent complementizer).

A similar scenario has been proposed for temporal *while*-clauses in English (> *ða hwil ðe* ‘the time that’; Trotta & Seppännen 1998).

(27)	[CP-relative	**die**	**wîl** [C°	**daz**]	fiwer (...)	under	erden	bran],	
		the.ACC	while.ACC	COMP	fire	under	earth	burned	
	[CP-matrix	**die**	**wîl**	schuof	der	soldan	werlicher	liut	genuoc
		the.ACC	while.ACC	made	the	sultan	armed	people	enough
		dringen	durch	daz	luoc] ...,				
		get-out	through	the.ACC	hole				

‘While a fire burned under the earth, during that time the sultan planned for armed people to surge through the hole ...’
(Middle High German; adapted from Eberhardt & Axel-Tober 2023, 278)

Alternatively, noun-based adverbial complementizers can be hypothesized to originate from headed relative clauses. This is the more traditional explanation that was reconstructed in a formalized way by Weiß (2019, 2020, and 2021). The starting point in this scenario is constructions as represented by the Middle High German examples in (28a, b), where there is a complex DP containing a relative clause introduced by the complementizer *dass* ‘that’ (or *so* ‘so’ and *und* ‘and’, see Weiß 2019, 2020, 2021 for details), whereby the relative complementizer could be omitted for stylistic reasons (28c, d).

Additionally, the definite article could be reduced to a clitic (28c) and eventually disappeared altogether (28d).

(28) a. **alldiewil** **das** ich uwer pflegen sol
all.the.while COMP I you.GEN care shall
‘as long as I shall care for you’

b. **dwil** **das** er kint was
the.while COMP he child was
‘while he was a child’

c. **Dwil** ich off ertrich on sunde nit enmocht gewesen
the.while I on earth without sin not NEG.could been
‘because I could not live on earth without sin’

d. di here cristenhait … sal loben … **Wile** ummer diese werlt gestet
the noble Christianity … shall praise … while always this world persists
‘the noble Christianity has to praise, as long as this world exists’
(Middle High German; Weiß 2019, 522-524)

In utterances like (28d), hearers can get the impression that it is the lexeme *weil* that introduces the subordinated clause. With the reanalysis of the noun as a complementizer, the former relative clause was simultaneously reanalyzed as a temporal clause. (29) sketches the structural development of *weil* from noun to complementizer. In a more theoretical frame, the reanalysis of *weil* as a complementizer has the effect that the lexeme is inserted earlier in the structure (following the so-called *Early Merge Principle* proposed in Weiß 2019).

(29) [$_{\text{DP}}$ d‘ [$_{\text{N}}$ wîle [$_{\text{CP}}$ Ø …]]] → [$_{\text{CP}}$ wîle …]

As said above, the development of *weil* allows for two explanations, which is rather the exception, but not unique in German. There are other languages where temporal clauses develop from internally headed relative clauses. Diessel (2019, 107) presents an example from Jamsay, a Dogon language spoken in Mali. Other cases, however, unambiguously follow the one or the other way.

4.2.3. Source constructions with prepositions

Headed relative clauses often constitute the source structure where prepositions or greater parts of a prepositional phrase grammaticalize into adverbial complementizers. An example to illustrate this developmental path is the German adverbial complementizer *seit* ‘since’ and its variant *seitdem* lit. ‘since that.DAT’ (Weiß 2019, 525ff.; 2021, 9ff.). In the source structure, the preposition took a demonstrative pronoun as its complement, and the demonstrative pronoun in turn embedded a relative clause (see 30a) (cf. also Axel-Tober 2012 for further Old and Middle High German examples). As a stylistic option, the demonstrative pronoun as well as the relative clause complementizer could be omitted (cf. 30b, c):

(30) a. **sît** **dem** sîn vreide sî ze wege
since that.DAT his delight be to way
'since his delight was on the way'

b. **sit** **daz** Kriemhilde ze wîbe gewan / Sîfrit min sune
since that.DAT Kriemhild to wife won Siegfried my son
'since Siegfried, my son, won Kriemhield as wife'

c. **sît** ez nieman sach
since it no one saw
'since no one saw it'
(Middle High German; adapted from Weiß 2019, 526f.)

(31) sketches the structural change associated with the reanalysis of *seit(dem)* as a complementizer.

(31) [$_{PP}$ sît (dem) [$_{CP}$ ...]] → [$_{CP}$ [$_{C°}$ sît(dem)] ...]

4.3. The emergence of asyndetic adverbial clauses

A further type of adverbial clause whose development has been hypothesized to involve a change from parataxis to hypotaxis is clauses without complementizers. Notably so-called 'asyndetic' conditionals with conditional or conditional-like interpretations were already discussed in Neogrammarian approaches on German (e.g., Behaghel 1928, 637) and by Otto Jespersen for English (Jespersen 1940). This clause type is attested in many Germanic languages (including their earliest stages, e.g., Axel-Tober & Wöllstein 2009). In Germanic, asyndetic adverbial clauses show verb-first order which is typical for polar interrogatives, see (32).

(32) **Scheint** die Sonne, gehen wir baden
shines the.NOM sun go.1PL we bathe
'If the sun is shining, we will go for a swim.'

This phenomenon can thus also be seen as a reflex of a general cross-linguistic affinity between interrogatives and conditionals. For example, there are also languages where interrogative particles and/or suffixes are used as conditional markers (see Harris & Campbell 1995, 294-310 for more examples).
Since the first sentence (the protasis) has the form of an independent interrogative sentence, the emergence of asyndetic conditionals is traditionally explained as going back to source structures as in (33) where the first sentence (= S1) is indeed a question. Such a diachronic scenario could be seen as a further example of Givón's (1979) claim that complex structures evolve through the syntacticization of discourse.

(33) A: Scheint die Sonne? = S1
shines the.NOM sun
B: Ja.
yes

A: So/Dann gehen wir baden. = S2
so/ then go.1PL we bathe
'A: Is the sun shining? B. Yes. A: Then, we'll go for a swim.'
(Based on van den Nest 2010, 95)

Harris & Campbell (1995, 294-310), however, object that the synchronic formal overlap between interrogatives and some types of subordinate clauses should not be seen as evidence for a diachronic development from one sentence type into the other, but probably results from the non-assertional character of questions. This might have led to the analogical extension of question-marking devices into non-assertional subordinate clauses. As to the Germanic verb-first subtype, the traditional parataxis-to-hypotaxis scenario has been confronted with empirical counterevidence. Van den Nest (2010), a proponent of this theory in general, points out to the fact that the present subjunctive was used frequently in the protasis in Old English, whereas independent polar questions were only possible with indicative. Furthermore, Axel-Tober & Wöllstein (2009) and Reis & Wöllstein (2010) provide many empirical and theoretical arguments that verb-first conditionals in historical and present-day German are not truly embedded, especially since they are not placed in the prefield of the main clause, but adjoined to a verb-first apodosis. Verb-first conditionals can also be used in verb-third configurations with a verb-second apodosis (*Sollte die Sonne herauskommen, wir würden an den Strand gehen* 'Should the sun come out, we would go to the beach') and are largely banned from post-posed and apodosis-internal placements. Unlike canonical syndetic conditionals, they cannot occur within the scope of apodosis-internal operators and are always realized with a separate focus-background structure.
To conclude, even though the emergence of asyndetic adverbial clauses seems to be a clear case of parataxis-to-hypotaxis evolution at first glance, much research on the Germanic verb-first conditionals has not been able to confirm this type of hypothetical route. Neither is there empirical support that the source structure was really one with an independent interrogative, nor does the target structure show the typical characteristics of a fully hypotactic construction.

5. Conclusion

The parataxis-to-hypotaxis hypothesis has figured prominently in many approaches to the evolution of complex syntactic structures from the 19th onwards. However, if one looks at the developmental steps of specific types of subordinate clauses in individual languages, the empirical support for the respective hypothetical routes is often weak. In their attested histories, the source constructions were most often already hypotactic in the first place. Notably relative constructions are major cross-linguistic sources in the development of complement clauses and adverbial clauses. Also, we find many cases of renewal of new relative markers based on older relative structures. In some cases, it could be argued that there was a diachronic development from weaker to stronger hypotactic structures.

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